

Math AA HL at KCA - Chapter 5 Notes

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Contents

1	Forms of Complex Numbers and Operations	3
1.1	Cartesian Form	3
1.2	Modulus-Argument Form	4
1.3	Exponential/Euler's Form	5
2	The Argand Diagram	6
3	Complex Roots	7

1 Forms of Complex Numbers and Operations

1.1 Cartesian Form

$$z = a + bi$$

where

- $a = \operatorname{Re}(z)$
- $b = \operatorname{Im}(z)$

The complex conjugate of z will then be

$$z^* = a - bi$$

The addition and subtracting of complex numbers in the Cartesian form are done by adding/subtracting the real components and the imaginary components respectively.

Division involves multiplying by the **complex conjugate** of the denominator throughout

$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)}$$

this is so that the imaginary component will be canceled out (similar to rationalizing denominators)

1.2 Modulus-Argument Form

$$z = r (\cos \theta + i \sin \theta)$$

where

- $r = |z|$ = modulus; the straight line distance from the origin to z on the Argand diagram
- $\theta = \arg(z)$ = argument; the angle measured counter-clockwise from the positive x -axis.

Operations on moduli and arguments

Suppose z and w are two complex numbers, then

$$|zw| = |z||w|$$
$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$$

$$\arg(zw) = \arg(z) + \arg(w)$$
$$\arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w)$$

Operations on modulus-argument

Multiplication:

$$(r_1(\cos \alpha + i \sin \alpha)) (r_2(\cos \beta + i \sin \beta)) = r_1 r_2 (\cos(\alpha + \beta) + i \sin(\alpha + \beta))$$

Division:

$$\frac{r_1 (\cos \alpha + i \sin \alpha)}{r_2 (\cos \beta + i \sin \beta)} = \frac{r_1}{r_2} (\cos(\alpha - \beta) + i \sin(\alpha - \beta))$$

1.3 Exponential/Euler's Form

$$z = re^{i\theta}$$

where

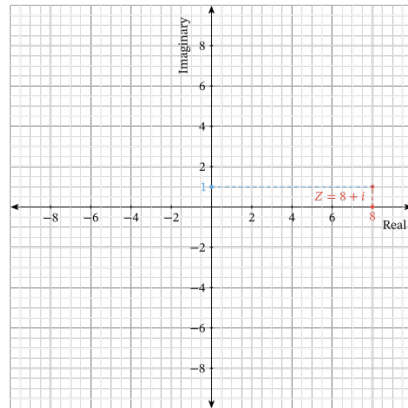
- $r = |z|$
- $\theta = \arg(z)$

Multiplication and division:

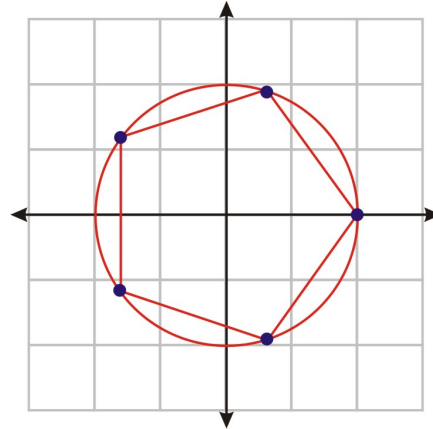
$$zw = r_1 r_2 e^{i(\alpha+\beta)}$$

$$\frac{z}{w} = \frac{r_1}{r_2} e^{i(\alpha-\beta)}$$

2 The Argand Diagram



(a) The Argand diagram



(b) 5th roots

Key points

1. For a complex number $z = a + bi$
 - (a) $|z| = \sqrt{a^2 + b^2}$
 - (b) $\arg(z) = \arctan(\frac{b}{a})$, but a further step of using other information such as the quadrant is required to determine the argument, since $\tan(\theta) = \tan(\pi + \theta)$.
2. The real axis is a line of symmetry for the complex solutions of a polynomial equation; this is because complex roots are **pairwise conjugate**.
3. The n -th roots of a complex number **have the same modulus** and **equally divide a full turn**. This leads to implications such as
 - (a) The roots join to form a circle with a radius equal to their mutual modulus.
 - (b) By joining the edges, a regular n -sided polygon is formed.
 - (c) The area of the regular polygon formed by the n -th root of a complex number with modulus r is given by $\frac{n}{2} \sqrt[n]{r^2} \sin(\frac{2\pi}{n})$

3 Complex Roots

Complex roots of a polynomial are pairwise conjugates. I.e. If one root is $z_1 = a + bi$, then there must exist another such that $z_2 = a - bi$.